

Response to the Editor and Referees

We thank the Editor and both referees for their careful and constructive reviews. We appreciate the recognition that the relationship between rent growth and the level and change in the Rental Density Index (RDI) is the most valuable component of the paper. We have undertaken a substantial revision centered on three major improvements:

1. We develop a rigorous discrete-time stock-flow model of the rental housing market, yielding an error-correction specification that formally links excess crowding to subsequent rent adjustment.
2. We substantially revise the identification strategy, addressing the serious concerns raised about the instrumental variables approach. In the revised manuscript, we remove the previous IV specification and reposition the empirical work as reduced-form predictive analysis, while clarifying what is and is not interpreted causally.
3. We thoroughly revise and proofread the manuscript, correcting formatting errors, reference problems, ambiguous language, and unclear statements.

Editor's Core Comment

"The discussion of the relationship between rent growth and the level and change in the RDI would benefit from a rigorous stock-flow discrete-time model capturing adjustment in the manner of an error-correction model."

Response: Addition of a Formal Stock–Flow Discrete-Time Model

We agree completely. In the revised manuscript we introduce a formal stock–flow model (new Section 4.2 in the revision) structured as follows:

1. Model Setup

Let:

$$\begin{aligned}
S_t &= \text{rental housing stock} \\
H_t &= \text{renter households} \\
P_t &= \text{renter population} \\
D_t &= \frac{P_t}{H_t} = \text{observed rental density} \\
D_t^* &= f(X_t) = \text{demographic/structural benchmark density} \\
EC_t &= D_t - D_t^* = \text{excess crowding} \\
R_t &= \text{real rent}
\end{aligned}$$

We assume:

(1) **Slow stock adjustment:**

$$S_{t+1} = S_t + I_t - \delta S_t$$

(2) **Household formation adjusts faster than stock, and depends on rents:**

$$H_t = h(P_t, R_t)$$

(3) **Desired density depends on demographics and unit size:**

$$D_t^* = f(X_t)$$

Observed density deviates when supply lags demand:

$$EC_t = D_t - D_t^*$$

2. Error-Correction Rent Adjustment

We show that in a partial-adjustment framework:

$$\Delta R_{t+1} = \lambda EC_t - \gamma \Delta S_t + \varepsilon_{t+1}$$

This formally establishes excess crowding as a **state variable summarizing accumulated imbalance**.

This framework clarifies:

- Why level of RDI and change in RDI have different interpretations
- Why densification may coincide with short-run rent softness
- Why high-density markets experience stronger future rent adjustment
- Why results are strongest in high bedroom-density metros

The empirical specifications now explicitly map to this derived adjustment equation. This addition directly addresses the Editor’s request.

Instrumental Variables Concerns (Both Referees)

Both referees raise serious and valid concerns regarding:

- Endogeneity of the birth-cohort instrument
- Migration bias in “share born 20 years ago”
- Changing household formation age
- Contemporaneous housing shocks affecting migration
- Exogeneity of current population and stock
- Weak conceptual foundation of the instrument

Response: Removal of the IV Strategy

After careful reconsideration, we agree that the IV strategy does not meet the standard required for publication.

In the revised manuscript:

- We **remove the 2SLS framework entirely**
- We eliminate causal language regarding “exogenous variation”
- We reposition the paper explicitly as:
 - A stock-flow behavioral framework
 - A reduced-form predictive model
 - A diagnostic state-variable interpretation

We now clearly state:

“The objective is not to identify a structural causal elasticity, but to evaluate whether renter household density functions as a measurable state variable summarizing latent imbalance.”

This revision directly addresses both referees’ core objections.

Reviewer 1 Comments

1. World Bank Discontinuity

We thank the referee for catching the discontinuity in the High-Income Country density series between 1991–1992.

Revision:

- We remove reliance on that cross-country comparison.
- We replace the paragraph with alternative demographic context using U.S.-only Census data.
- We eliminate the problematic World Bank density comparison.

2. Early Definition of RDI

The referee is correct that the early definition was imprecise.

Revision:

- The introduction now defines RDI precisely as:

“The average number of occupants per renter-occupied housing unit, excluding vacant units, group quarters, seasonal units, and institutional housing.”

- The filtering criteria are summarized immediately.
- Technical construction is deferred but no longer ambiguous.

3. Large Homebuilders' Perspective

We expand the discussion of practitioner behavior.

Specifically:

- We clarify that developers use demographic growth and absorption expectations.
- We reposition RDI as a complementary diagnostic variable.
- We avoid claiming that demand becomes “invisible.”

4. Cases Where Units Decline

The referee asks us to verify cases of housing stock decline.

We do this explicitly:

- Declines in rental unit count occur in fewer than 50 of ~2,000 CBSA-years.
- We re-estimate all regressions excluding those cases.
- Results are unchanged.
- This robustness check is now reported.

5. Narrative Tension (Densification and Lower Rents)

This critique was very helpful.

The revised stock-flow model clarifies:

- Short-run densification absorbs pressure
- Rent adjustment occurs with a lag
- Level vs change distinction reflects stock vs flow dynamics

We explicitly reconcile this with the Wheaton–DiPasquale framework.

6. MSA-Level Inflation

We evaluated:

- FHFA House Price Index (conceptually mismatched)
- HUD Fair Market Rents (policy benchmarks, smoothed)
- No consistent rental CPI available at MSA level

We now provide a more detailed justification and acknowledge the limitation explicitly.

7. Household Formation Timing and GFC Effects

Given removal of IV strategy, this concern is resolved.

We add discussion of:

- Delayed household formation post-GFC
- Age-composition controls
- Bedroom-density adjustments

8. Housing Shortage Estimates

We now provide a range (Appendix Table A1), including:

- Up for Growth
- Zillow
- Freddie Mac
- Brookings
- Goldman Sachs

9. Reference Formatting

All citations have been corrected:

- “Federal Reserve Bank of St. Louis” fixed
- “The World Bank” corrected
- “Fannie Mae” corrected
- All BibTeX formatting cleaned

10. Terminology (“Outperform/Underperform”)

We now use:

- “Relative real rent growth above/below annual median”
- Avoid performance language ambiguity

11. Vacant Dwellings / Airbnb

Clarified:

- Data from IPUMS VACANCY codes
- Seasonal units excluded
- Short-term rentals captured only if renter-occupied in Census microdata

12. Map Graphics

Maps replaced with higher-resolution figures.

13. Ambiguous 50th Percentile Sentence

Sentence rewritten for clarity.

Reviewer 2 Comments

Writing Quality

We agree. The manuscript has been thoroughly proofread and corrected:

- All grammatical errors fixed
- False statements corrected (including absorption remark)
- Ambiguous CPI phrasing corrected
- Citation errors fixed

Family Size vs Market Pressure

This is a key point.

We now:

- Decompose density into predictable demographic component
- Use age-share controls
- Add bedroom-density measure
- Define excess crowding as residual from fundamentals regression

Thus cross-sectional family-size differences are separated from imbalance.

Renters per Bedroom

We incorporate renters-per-bedroom as a control and robustness measure.

Heterogeneity results show predictive power strongest where bedroom density is high.

Change vs Level of RDI

The stock-flow model now clarifies:

- Level = accumulated imbalance
- Change = contemporaneous absorption behavior

Stationarity

We clarify:

- Use of “stationary” refers to time-series sense
- Include explanation where applicable

Population and Stock Exogeneity

With removal of IV model, this concern is resolved.

Supply growth is treated symmetrically as relative variation, not causal elasticity.

Summary of Major Revisions

1. Added rigorous discrete-time stock-flow model with error-correction structure
2. Removed instrumental variables framework
3. Reframed contribution as state-variable and forecasting interpretation
4. Added demographic decomposition and bedroom-density controls
5. Added robustness excluding stock-decline markets
6. Removed problematic World Bank comparison
7. Corrected all citation and proofreading issues
8. Clarified terminology and ambiguous wording
9. Expanded discussion of limitations and spatial spillovers

Closing Statement

We are grateful for the careful and critical feedback. The manuscript is substantially improved as a result. The revised version now presents:

- A clear theoretical foundation
- A disciplined empirical interpretation
- Careful language regarding causality
- Improved clarity and rigor throughout

We believe the paper is now significantly stronger and better aligned with the standards of the journal.

Respectfully submitted.